

ENGINEERING CHEMISTRY LAB RECORD WRITING

Colour-coded guide for the complete laboratory manual

RIGHT SIDE — MAIN RECORD MATTER

Write the experiment number/date, title, aim, apparatus, reagents, principle/theory, equations, diagrams, procedure and other required record matter on the RIGHT SIDE.

LEFT SIDE — OBSERVATIONS & CALCULATIONS

Write observation tables/readings and all calculations on the LEFT SIDE. Calculation/formula work belongs with the observations on the LEFT.

RESULT — RIGHT SIDE ONLY & AT THE END

Write the final RESULT at the end of each experiment and on the RIGHT SIDE ONLY. Keep the result clearly separated from the calculations.

NOT REQUIRED — DO NOT WRITE

MODEL GRAPHS & MODEL DIAGRAMS · LEFT SIDE ONLY

POGIL ACTIVITY is NOT required to be written in the student record. PRECAUTIONS are also NOT required to be written in the student record. The diagrams are marked separately in grey.

Experiment no:

Date:

RIGHT SIDE - MAIN RECORD MATTER

Estimation of Hardness of Water by Complexometric method using EDTA

Aim: To estimate the total hardness of water by Complexometric method using EDTA.

Apparatus required:

1. Beaker
2. Burette
3. Pipette
4. Conical flask
5. Wash bottle

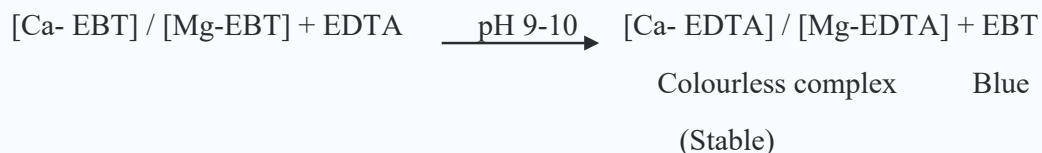
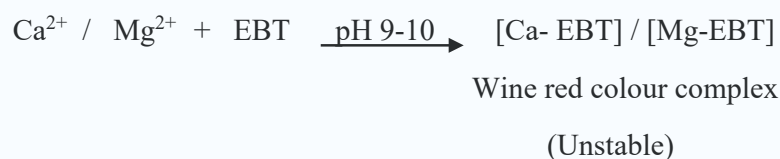
Reagents required:

1. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (Epsom salt)
2. Disodium salt of Ethylenediaminetetraacetic acid
3. Eriochrome Black-T (EBT) indicator
4. Buffer ($\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$)

Principle:

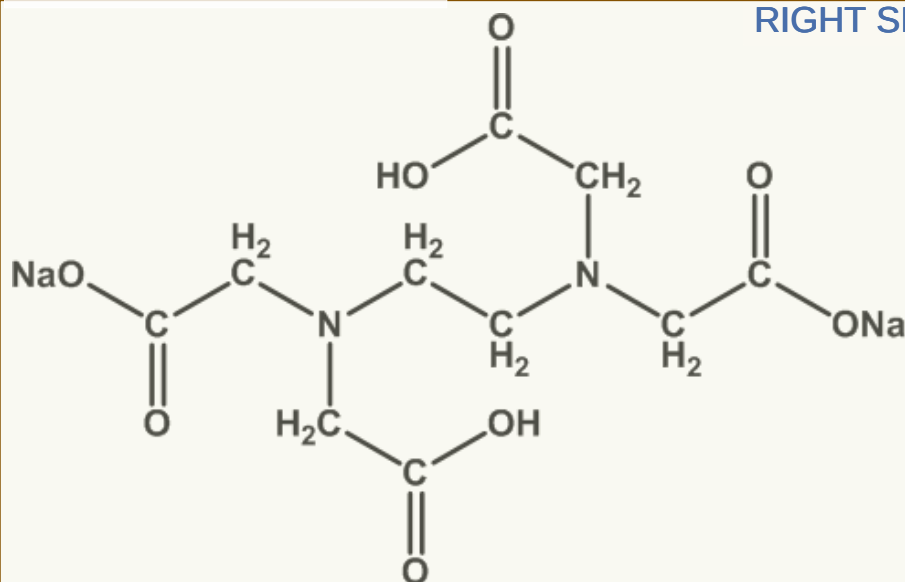
$\text{Ca}^{2+}(\text{aq})$ and $\text{Mg}^{2+}(\text{aq})$ ions that cause hardness in water can be determined by complexometric titration with a chelating agent, ethylenediaminetetraacetic acid (EDTA), usually in the form of disodium salt.

Addition of Eriochrome Black-T indicator to the hard water at pH range 9-10 forms wine red coloured unstable complex with Ca^{2+} and Mg^{2+} ions of the sample water. The unstable wine red colour complex forms a colourless stable complex with EDTA and liberates the blue coloured EBT indicator.



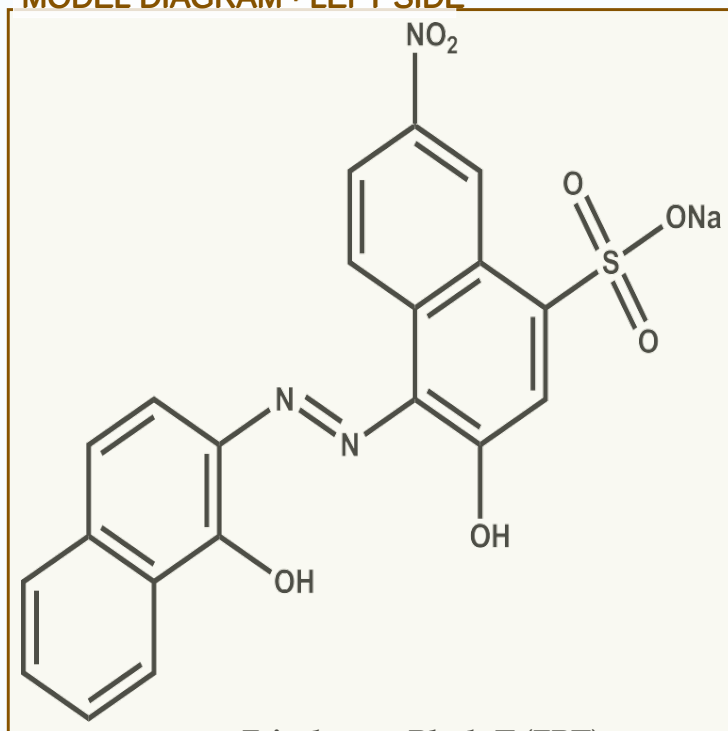
MODEL DIAGRAM · LEFT SIDE

RIGHT SIDE · MAIN RECORD MATTER



Disodium salt of Ethylenediaminetetraacetic acid

MODEL DIAGRAM · LEFT SIDE



Eriochrome Black-T (EBT)

Procedure:**Step- 1: Preparation of standard $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solution**

- Weigh the given $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ salt using weighing balance and transfer into a clean 100 ml volumetric flask.
- Dissolve the transferred salt in the distilled water and make up the solution up to the mark.
- Shake the volumetric flask for a while to get uniform concentration.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Weight of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ salt + weighing bottle (W_1) =

Weight of empty weighing bottle (W_2). =

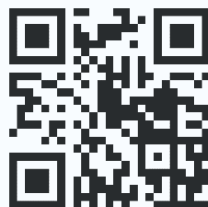
Weight of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ salt transferred ($W_1 - W_2$). =

$$\text{Normality of } \text{MgSO}_4 \cdot 7\text{H}_2\text{O} = \frac{\text{Weight of salt}}{\text{Equivalent weight of salt}} \times \frac{1000}{100}$$

=

=

(Equivalent Weight of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ is 123)

Step- 2: Standardization of EDTA solution**RIGHT SIDE · PROCEDURE**

- Pipette out 20ml of Standard $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solution into a clean 250 ml conical flask.
- Add 2 ml of buffer ($\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$) to maintain pH between 9-10.
- Add 2-3 drops of Eriochrome Black-T indicator, the solution then turns to wine red.
- Fill the burette with EDTA solution and adjust to zero, then fix it to the burette stand.
- Now titrate the above $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solution against EDTA solution till the wine-red solution turns to blue which is the end point of the titration.
- Note down the burette reading and repeat the titration for concurrent values.

$$N_1 V_1 = N_2 V_2$$

LEFT SIDE · OBSERVATIONS & CALCULATIONS

S.No	Volume of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solution (ml)	Burette reading (ml)		Volume of EDTA solution run down (ml)
		Initial	Final	
1	20			
2	20			
3	20			

N_1 = Normality of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solution =

V_1 = Volume of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ solution =

N_2 = Normality of EDTA solution =

V_2 = Volume of EDTA solution =

$$\text{Normality of EDTA solution } (N_2) = \frac{N_1 V_1}{V_2}$$

=

=

RIGHT SIDE · PROCEDURE

Step 3: Estimation of total hardness of the given water samples



- Pipette out 20 ml of water sample-1 into a clean 250 ml conical flask.
- Add 2 ml of buffer ($\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$) to maintain pH between 9–10.
- Add 2–3 drops of Eriochrome Black-T indicator then solution turns to wine red in colour.
- Fill the burette with standardized EDTA solution and adjust to zero then fix it into the burette stand.
- Now titrate the above water sample against EDTA solution till the wine-red solution turns to blue which is the end point of the titration.
- Note down the burette reading and repeat the titration for concurrent values.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Observations and Calculations:

	Volume of water sample (ml)	Burette reading		Volume of EDTA solution run down (ml)
		Initial	Final	
Sample Code -----				
				(V ₃)
Sample Code -----				
				(V ₄)

Normality of EDTA = (Given)

Total hardness expressed as CaCO₃ equivalents

(*Equivalent weight of CaCO₃ is 50)

$$\text{Normality of EDTA} \times \text{Volume of EDTA} \times 50 \times 1000 = \frac{\text{Volume of water sample}}{\text{Volume of water sample}}$$

$$\text{Water Sample} \text{---} = \frac{N_2 \times \text{Volume of EDTA}(V_3) \times 50 \times 1000}{\text{Volume of water sample}}$$

=

=

$$\text{Water Sample} \text{---} = \frac{N_2 \times \text{Volume of EDTA}(V_4) \times 50 \times 1000}{\text{Volume of water sample}}$$

=

= PRECAUTIONS · NOT REQUIRED

Precautions:

1. All the glassware must be washed with distilled water
2. All the reagents must be prepared freshly with distilled water

RESULT · RIGHT SIDE ONLY

Amount of total hardness present in the given	Reported	Given	% Error
Water sample ____ in ppm			
Water sample ____ in ppm			

Result:

$$\% \text{Error} = \frac{|\text{Given} - \text{Reported}|}{\text{Given}} \times 100$$

Engineering Chemistry Lab

POGIL ACTIVITY

I. Title: Estimation of hardness of water by complexometric method using EDTA

II. Learning Objectives:

- i. To determine the hardness of water samples by complexometric titration.
- ii. To compare the amount of hardness in the given water samples.

III. Critical thinking Questions:

1. Name the titration involved in this experiment?
 - a) Precipitation titration b) Complexometric titration c) Acid -Base titration d) Redox titration
2. What is the colour change at the end point?
 - a) blue to wine red b) wine red to blue c) red to blue d) blue to pink
3. The type and composition of the buffer used in the titration is.
 - a) Acidic buffer (HCl + NH_4Cl) b) Basic buffer ($\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$)
 - c) Acidic buffer (HCl + NaOH) d) Basic buffer (NaOH+NaCl)
4. What is the hardness range of hard water?
 - a) 50 – 100 ppm b) 100 – 150 ppm c) 150 – 300 ppm d) 300 above
5. At what favorable pH range, complexes are formed.
 - a) 6-7 b) 7-8 c) 9-10 d) 8-9
6. Write the unstable complex formed in the reactions -----.
7. Temporary hardness is estimated by----- method.
8. Hardness of water is expressed in ----- units.
9. -----is the primary standard used in the hardness experiment.
10. The indicator used in the determination of hardness of water by EDTA method is-----.
11. Why natural water should not be fed into boilers?
12. Soft water is not fit for drinking. Justify
13. Blue color is formed at the end point. Give reason
14. Suggest another method for estimating the amount of hardness in water sample.
15. What are the permissible and desirable limits of hard water?

6. Research Applications:

What is the information obtained by performing this experiment and its utility in various fields of water technology? (200 words)

Hints:

- 1) What happens if hard water is used in various industries?
- 2) With special reference to steam generation in boilers.

POGIL ACTIVITY

Engineering Chemistry Lab

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE - MAIN RECORD MATTER

Determination of chloride content in the given sample water by Argentometric method

Aim: To determine the chloride content in the given water sample using silver nitrate solution.

Apparatus required:

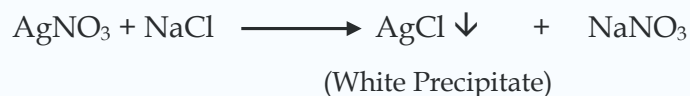
1. Weighing bottle
2. Volumetric flask
3. Beaker
4. Burette
5. Pipette
6. Conical flask
7. Wash bottle

Reagents required:

1. Sodium Chloride (NaCl)
2. Potassium Chromate (K_2CrO_4)
3. Silver Nitrate ($AgNO_3$)

Principle:

Chloride ions are precipitated by silver ions which combine with chromate ions to form brick red silver chromate. Silver chloride formation is fast and irreversible and on the other hand formation of silver chromate is slow and reversible. Therefore, till all the chloride ions are exhausted silver ions do not give a stable silver chromate.



Procedure:

RIGHT SIDE · PROCEDURE

Step 1: Preparation of standard NaCl solution



- Weigh accurately the given NaCl salt using the weighing balance and transfer into a clean 100 ml volumetric flask.
- Dissolve it in distilled water and make up the solution up to the mark.
- Shake well for uniform concentration.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Weight of NaCl salt + weighing bottle (W_1) =

Weight of empty weighing bottle (W_2) =

Weight of NaCl salt transferred ($W_1 - W_2$) =

$$\text{Normality of NaCl solution} = \frac{\text{Weight of salt}}{\text{Equivalent weight of salt}} \times \frac{1000}{100}$$

=

=

(Equivalent Weight of NaCl is 58.5)

Step-2B: Standardization of AgNO_3 solution

RIGHT SIDE · PROCEDURE



- Pipette out 10 ml of standard NaCl solution into a clean 250 ml conical flask.
- Add 2 or 3 drops of potassium chromate solution, the solution turns yellow.
- Fill the burette with AgNO_3 solution and adjust to zero, then fix it to the burette stand.
- Now titrate against AgNO_3 solution till the appearance of brick red colour which is the end point.
- Note down the burette reading and repeat the titration for concurrent values.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

S.No	Volume of NaCl solution (ml)	Burette reading (ml)		Volume of AgNO ₃ solution run down (ml)
		Initial	Final	
1	10			
2	10			
3	10			
4	10			

N_1 – Normality of NaCl =

V_1 – Volume of NaCl =

N_2 – Normality of AgNO₃ =

V_2 – Volume of AgNO₃ =

$$N_1 V_1 = N_2 V_2$$

$$\text{Normality of AgNO}_3 \text{ solution } (N_2) = \frac{N_1 V_1}{V_2}$$

=

=

Step 3: Estimation of chloride in the given water sample:

RIGHT SIDE · PROCEDURE



- Pipette out 10 ml of water sample into a clean 250 ml conical flask.
- Add 2 or 3 drops of potassium chromate solution, the solution turns yellow.
- Fill the burette with standardized AgNO₃ solution and adjust to zero, then fix it to the burette stand.
- Now titrate against AgNO₃ solution till the appearance of brick red colour which is the end point.
- Note down the burette reading and repeat the titration for concurrent values.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Observations and Calculations:

	Volume of water sample (ml)	Burette reading (ml)		Volume of AgNO ₃ solution run down (ml)
		Initial	Final	
Sample Code _____				
				(V3)
Sample Code _____				
				(V4)

Normality of AgNO₃ solution =

Chloride content in given

$$\text{Water Sample } \underline{\hspace{2cm}} = \frac{\text{Normality of AgNO}_3 \times \text{Volume of AgNO}_3 \text{ (V3)} \times 35.5 \times 1000}{\text{Volume of water sample}}$$

(*Equivalent weight of chloride ion is 35.5)

=

=

$$\text{Water Sample } \underline{\hspace{2cm}} = \frac{\text{Normality of AgNO}_3 \times \text{Volume of AgNO}_3 \text{ (V4)} \times 35.5 \times 1000}{\text{Volume of water sample}}$$

(*Equivalent weight of chloride ion is 35.5)

=

=

PRECAUTIONS · NOT REQUIRED

Precautions:

1. All the apparatus must be washed with distilled water only. **PRECAUTIONS : NOT REQUIRED** because tap water contains chloride ions.
2. All the reagents must be prepared with distilled water only.

Result:

RESULT · RIGHT SIDE ONLY

Amount of chloride content present in the given	Reported	Given	% Error
Water sample _____ in ppm			
Water sample _____ in ppm			

$$\% \text{ Error} = \frac{|\text{Given} - \text{Reported}|}{\text{Given}} \times 100$$

Engineering Chemistry Lab

POGIL ACTIVITY

I. Title: Estimation of chloride content in water using silver nitrate

II. Learning Objectives:

- To find the impurities like chloride in the given water sample.
- To understand the purity of the water samples.

III. Critical thinking Questions:

1. What type of titration is chloride ion estimation?
 - a) Precipitation titration
 - b) Complexometric titration
 - c) Redox titration
 - d) Acid-base titration
2. What are the sources of chloride content in water sample?
 - a) NaCl
 - b) $MgCl_2$
 - c) $CaCl_2$
 - d) All the above
3. Indicator used in the titration.
 - a) Methyl orange
 - b) Potassium Chromate
 - c) Methyl red
 - d) Eriochrome black T
4. Colour change at the end point.
 - a) Wine red to blue
 - b) pale yellow to brick red
 - c) brick red to pale yellow
 - d) colourless to pale pink
5. Formation of silver chloride is
 - a) fast and irreversible
 - b) slow and irreversible
 - c) slow and reversible
 - d) fast and reversible
6. Name the stable product formed in the reaction _____
7. Desirable limit of chloride ion in drinking water _____
8. _____ is the primary standard solution used in this experiment
9. The concentration of chloride content is expressed in terms of _____
10. Name the chloride salts that cause permanent hardness _____
11. What is the industrial importance of this experiment?
12. Why the presence of chloride in drinking water is not recommended?
13. Discuss the advantages and disadvantages of adding liquid chlorine in swimming pools.
14. Name the methods used to remove chloride content in drinking water.
15. What is available chlorine in bleaching powder?

IV. Research Applications:

Write the applications for chloride ion determination?

Hints:

- 1) Industrial applications
- 2) Swimming pools & drinking water

POGIL ACTIVITY

Engineering Chemistry Lab

POGIL ACTIVITY

Experiment no:

Date:

Determination of Iron using Potassium Dichromate

RIGHT SIDE - MAIN RECORD MATTER

Aim: To estimate the amount of iron (ferrous), present in the given solution using a standard solution of potassium dichromate ($K_2Cr_2O_7$).

Apparatus:

1. Burette
2. Conical Flask
3. Beakers
4. Weighing Bottle
5. Weighing Balance
6. Measuring jar
7. Wash bottle.

Chemicals and Reagents:

1. $K_2Cr_2O_7$ Salt
2. Acid mixture (10ml H_2SO_4 + 2ml H_3PO_4)
3. $FeSO_4$ solution
4. Diphenylamine (indicator).

Principle:

Ferrous iron (Fe^{2+}) is oxidized to ferric iron (Fe^{3+}) by potassium dichromate in acidic medium. The completion of oxidation reaction is indicated by the appearance of the bluish violet color of diphenylamine, which is used as an internal indicator.



Overall reaction:



This is a redox reaction. $K_2Cr_2O_7$ is acting as an oxidizing agent and Fe (II) is acting as reducing agent. So, one equivalent weight of potassium dichromate oxidizes six equivalents of iron. Moreover, the mechanism of the oxidation of diphenylamine (indicator) to give a bluish violet colour at the end point. To avoid the premature end-point phosphoric acid is also added along with the mineral acid (H_2SO_4). Addition of phosphoric acid allows dichromate to oxidize Fe(II) first in preference to the indicator. After complete oxidation of Fe(II) by dichromate ion, then only indicator gets oxidized.

Procedure:

STEP-1: Preparation of standard potassium dichromate solution



- Weigh accurately (~ 0.25 g) the given pure sample of potassium dichromate and transfer it into 100 ml standard flask with a funnel.
- Dissolve it in small quantity of distilled water and make up the solution up to the mark using distilled water. Shake well for uniform concentration.
- Calculate the normality of potassium dichromate solution.

Observations and Calculations:

- Weight of $K_2Cr_2O_7$ salt + weighing bottle (W_1) =
- Weight of empty weighing bottle (W_2) =
- Weight of $K_2Cr_2O_7$ salt ($W_1 - W_2$) =

$$\text{Normality of } K_2Cr_2O_7 \text{ solution } (N_1) = \frac{\text{Weight of } K_2Cr_2O_7 \text{ salt}}{\text{Equivalent weight}} \times \frac{1000}{100}$$

=

$$\text{Normality of } K_2Cr_2O_7 \text{ solution } (N_1) =$$

(* Equivalent Weight of $K_2Cr_2O_7$ is 49)

Step-2: Estimation of Iron (Ferrous):

- Take 10 ml of given Fe(II) solution into a clean 250 ml conical flask.
- Add 10 ml of dil. H_2SO_4 , 2 ml of H_3PO_4 , 2-3 drops of diphenylamine indicator.
- Fill the burette with Std. potassium dichromate solution and adjust to zero, then fix it to the burette stand.
- Now titrate against $K_2Cr_2O_7$ solution till the appearance of bluish violet color via green colour.
- Note down the burette reading and repeat the titration for concurrent values.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

	Volume of Fe (II) Solution (ml) (V_2)	Burette reading (ml)		Volume of $K_2Cr_2O_7$ solution run down (ml) (V_1)
		Initial	Final	
Sample Code _____				

LEFT SIDE · OBSERVATIONS & CALCULATIONS N_1 – Normality of $K_2Cr_2O_7$

=

(From step-1)

 V_1 – Volume of $K_2Cr_2O_7$

=

 N_2 – Normality of Fe (II) solution

= ?

 V_2 – Volume of Fe (II) solution

=

$$N_1V_1 = N_2V_2$$

$$\text{Normality of Fe (II) solution } N_2 = \frac{N_1V_1}{V_2}$$

=

Strength of given Fe(II) solution = $N_2 \times \text{Eq. wt of Fe}^{2+}$

=

gm/L

(*Equivalent weight of Fe (II) is 55.85)

Result:**RESULT · RIGHT SIDE ONLY**

Amount of Ferrous ion present in the given solution	Reported	Given	% Error
Sample code _____			

POGIL ACTIVITY · NOT REQUIRED**POGIL ACTIVITY****I. Critical thinking Questions:**

- The estimation of Fe(II) using potassium dichromate is based on which type of reaction?
 - Precipitation
 - Acid–base neutralization
 - Oxidation–reduction
 - Complex formation

Engineering Chemistry Lab

2. In the dichromate method, Fe^{2+} is oxidized to Fe^{3+} by:
 - a) Potassium permanganate
 - b) Potassium dichromate
 - c) Potassium ferricyanide
 - d) Sodium thiosulphate
3. The indicator used in the titration of Fe(II) with $\text{K}_2\text{Cr}_2\text{O}_7$ is:
 - a) Methyl orange
 - b) Potassium ferricyanide
 - c) Diphenylamine
 - d) Phenolphthalein
4. The color of potassium dichromate solution is:
 - a) Colorless
 - b) Deep purple
 - c) Orange
 - d) Yellow
5. The reaction medium used during the titration of Fe(II) with dichromate is:
 - a) Basic medium
 - b) Acidic medium
 - c) Neutral medium
6. Why is sulfuric acid preferred over hydrochloric acid in the dichromate titration?
 - a) It gives better color change
 - b) HCl reacts with dichromate liberating chlorine gas
 - c) It prevents oxidation of Fe (II)
 - d) It is cheaper
7. The equivalent weight of potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) in redox reactions is:
 - a) 49
 - b) 98
 - c) 294
 - d) 7
8. In the redox reaction between Fe^{2+} and dichromate, the oxidation state of chromium changes from:
 - a) +6 to +3
 - b) +3 to +6
 - c) +2 to +6
 - d) +6 to +2
9. The endpoint in Fe(II) vs $\text{K}_2\text{Cr}_2\text{O}_7$ titration using diphenylamine is indicated by:
 - a) Colorless to red
 - b) Green to violet
 - c) Blue to green
 - d) Colorless to pink
10. Diphenylamine indicator changes color due to:
 - a) Formation of Fe^{3+} ions
 - b) Reduction of dichromate
 - c) Oxidation of diphenylamine
 - d) Formation of complex with Fe^{2+}
11. The purpose of adding phosphoric acid in some dichromate titrations is to:
 - a) Remove excess Fe^{3+} color
 - b) Form a complex with Fe^{3+} to sharpen the endpoint

- c) Neutralize the medium
- d) Increase oxidation potential
- 12. Define Dichrometry?
- 13. Define oxidation? Write the structure of diphenylamine and phosphoric acid?
- 14. What is the difference between ferric and ferrous ions?
- 15. What are the four types of titrations?
- 16. Why $K_2Cr_2O_7$ is strong oxidizing agent?
- 17. Why ferric ion is more stable than ferrous ion?

II. Research Applications:

- 1. Environmental Monitoring
 - a) Water quality analysis
 - b) Soil and sediment studies
 - c) Pollution studies
- 2. Biochemical and Microbiological Studies
 - a) Iron metabolism studies
 - b) Microbial iron reduction research
- 3. Corrosion Research
- 4. Geochemical and Mineral Studies
- 5. Industrial and Material Applications

Engineering Chemistry Lab

POGIL ACTIVITY

POGIL ACTIVITY

Experiment no:

Date:

Conductometric Titration of Strong Acid vs Strong Base

RIGHT SIDE : MAIN RECORD MATTER

Aim: To estimate the strength of HCl conductometrically using standard NaOH solution.

Apparatus required:

1. Conductivity meter
2. Conductivity cell
3. 10 ml micro burette
4. 250 ml conical flask
5. 100 ml beaker

Reagents required:

1. Oxalic acid
2. NaOH solution
3. HCl solution
4. Phenolphthalein Indicator

Principle:

The determination of equivalence point of a titration by this method is based upon the measurement of the conductance during titration. The conductance varies before and after the equivalence point. The reason for this is electrical conductance of solution depends upon the number of ions and their mobilities. A graph is drawn between conductance and volume of titrant added. Two straight lines are obtained. On extrapolation these lines intersect at a point. The point of intersection gives the end point and volume of titrant required for neutralization.



Procedure:

Step-1: Preparation of standard Oxalic acid solution



- Weigh accurately about 3.2 g of the given oxalic acid salt using the weighing balance and transfer into a clean 100 ml volumetric flask.
- Dissolve the transferred salt in distilled water and make up the solution up to the mark.
- Shake the volumetric flask for a while to get uniform concentration.

LEFT SIDE : OBSERVATIONS & CALCULATIONS

Weight of Oxalic acid salt + weighing bottle (W_1) =

Weight of empty weighing bottle (W_2) =

Weight of Oxalic acid salt transferred ($W_1 - W_2$) =

LEFT SIDE · CALCULATION

$$\text{Normality of Oxalic acid salt solution} = \frac{\text{Weight of salt}}{\text{Equivalent weight of salt}} \times \frac{1000}{100}$$

$$=$$

$$=$$

(Equivalent Weight of Oxalic acid is 63)

Step-2: Standardization of NaOH solution**RIGHT SIDE · PROCEDURE**

- Pipette out 20 ml of oxalic acid solution into a 250 ml conical flask.
- Add one or two drops of phenolphthalein indicator.
- Fill the burette with sodium hydroxide adjust to zero, then fix it to the burette stand.
- Now titrate the above oxalic acid solution against NaOH solution till the colourless solution turns to pale pink which is the end point of the titration.
- Note down the burette reading repeat the titration until concurrent readings are obtained.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Observations and Calculations:

S. No.	Volume of oxalic acid (V ₁ ml)	Burette reading(ml)		Volume of NaOH solution run down (v ₂ ml)
		Initial	Final	
1.	20			
2.	20			
3.	20			
4	20			

$$V_1 N_1 = V_2 N_2$$

Where V₁ -Volume of oxalic acid solution =

N₁ - Normality of oxalic acid solution =

V₂ - Volume of sodium hydroxide solution =

N₂ - Normality of sodium hydroxide solution = ?

$$N_2 = \frac{N_1 V_1}{V_2}$$

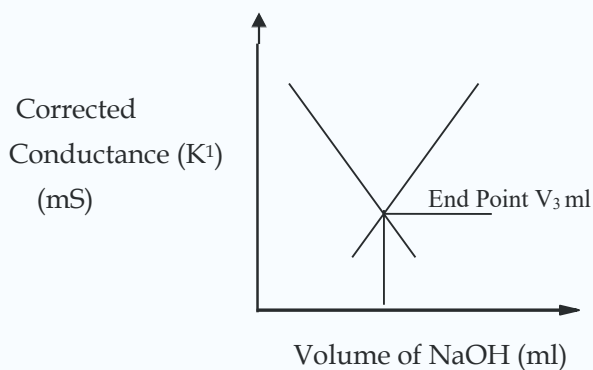
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Step 3: Estimation of HCl



- Make up the given unknown solution up to the mark with distilled water.
- Transfer 50 ml of unknown HCl solution with the help of 50 ml burette into a 100ml beaker.
- Immerse the conductivity cell in the HCl solution.
- Fill the micro burette with NaOH solution and fix it to the stand.
- Add 0.5 ml of NaOH from the micro burette into the beaker, stir well and note the conductance.
- Continue the addition of NaOH up to 10 ml in 0.5 interval.
- Conductance first decreases by the addition of NaOH and increases after the equivalence point is reached.
- Dilution can reduce the conductance of HCl solution. Corrected conductance (k^1) should be calculated to remove the influence of dilution factor in conductance (k).
- After the dilution (volume) correction to the conductance (k), plot the graph between corrected conductance (k^1) vs volume of NaOH.
- Draw a graph plotting the volume of NaOH on X-axis and corrected conductance (k^1) on Y-axis. Two straight lines are obtained. When the straight lines are extrapolated, they intersect at a point giving a V-shaped graph. The point of intersection (abscissa) is the end point that corresponds to the total volume of NaOH required for the complete neutralization of given unknown HCl. From that the strength of HCl can be calculated.



LEFT SIDE · OBSERVATIONS & CALCULATIONS

Observations and Calculations.

S.No	Volume of NaOH added in ml (V)	Conductance in mS At 25°C (k)	Corrected Conductance (k ^l) $\left[\frac{50+V}{50} \right] * k$
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			

MODEL GRAPH · LEFT SIDE

LEFT SIDE · CALCULATIONS

Determination of strength of hydrochloric acid:

$$V_1N_1 = V_2N_2$$

V_1 – Volume of sodium hydroxide solution = _____ ml (from graph)

N_1 – Normality of sodium hydroxide solution = _____ N (given)

V_2 – Volume of hydrochloric acid solution = 50 ml

N_2 – Normality of hydrochloric acid = ?

$$\text{Normality of hydrochloric acid } (N_2) = \frac{N_1V_1}{V_2}$$

=

=

Strength of HCl = Normality of HCl x Eq.Wt. of HCl

$$= N_2 \times 36.5$$

=

=

g/L

PRECAUTIONS · NOT REQUIRED

Precautions:

1. The electrodes must be immersed properly.
2. ~~The temperature of the solution should not change during the measurement.~~

RESULT · RIGHT SIDE ONLY

Result:

	Reported	Given	% Error
The strength of HCl present in the given solution in g/L			

MODEL DIAGRAM · LEFT SIDE

$$\% \text{ Error} = \frac{|\text{Given} - \text{Reported}|}{\text{Given}} \times 100$$

POGIL ACTIVITY

I. Title: Determination of the strength of HCl using standard NaOH conductometrically.

II. Learning Objectives:

- i. To learn the concept of conductance and the operation of Conductivity meter.
- ii. To monitor the total conductance of a solution during the titration and to determine the equivalence point.
- iii. To learn the role of conductance in determining the strength of the given solution.

III. Critical thinking Questions:

1. Units of conductance are
 - a) Ohms b) Moles / L c) Siemens d) cm^{-1}
2. Conductance depends upon
 - a) Temperature b) Mobility c) Concentration d) All
3. The shape of the graph obtained
 - a) X-shape b) V-shape c) Y-shape d) S-shape
4. The indicator used in the standardization of sodium hydroxide is
 - a) Methyl orange b) EBT c) Phenolphthalein d) Methyl red
5. The titrations in which endpoint is determined by conductance measurements are called _____
6. The platinum strips in a conductivity cell are coated with platinum black for _____
7. Calibration of an instrument is done to get _____
8. The conductance of the acid _____ with increase in volume before endpoint.
9. Solution used for calibration of conductivity meter is _____.
10. Why is AC current used in conductometric titration?
11. How temperature affects this titration?
12. Why the conductance range should be changed during the calibration?
13. Why does conductance of a solution not become zero at the end point?
14. Why the conductance of HCl is three times higher than NaOH though both are strong?

IV. Research Applications:

What is the information obtained by performing this experiment and its utility in various fields? (200 words)

Hints:

1. How conductance measurements are useful in chemical and biological fields?
2. Use of conductance measurements in various industries.

Engineering Chemistry Lab

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE - MAIN RECORD MATTER

Titration of Strong Acid Vs Strong Base by pH Metry

Aim: To determine the strength of unknown HCl solution using std NaOH by pH metry.

Apparatus:

1. pH meter
2. Combined glass electrode
3. 10 ml Micro burette
4. 100 ml Beaker

Reagents Required:

1. HCl solution
2. NaOH solution

Principle:

The determination of H^+ ion concentration depends on the fact that when a thin glass membrane is in contact with solutions of different H^+ ion concentrations on its two sides, a potential difference is developed across the glass membrane, depending on the concentration of the hydrogen ions in two solutions.

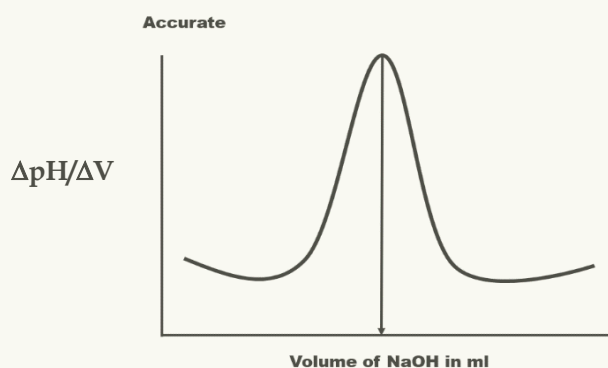
The determination of equivalence point of a titration by this method is based upon the measurement of the pH during titration. The pH varies before and after the equivalence point because the neutralization of acid and base is always accompanied by the change in concentration of H^+ and OH^- ions. A sudden change in pH is observed at the end point.



Estimation of HCl

- Make up the given unknown HCl solution up to the mark with distilled water.
- Transfer 50 ml of unknown HCl solution with the help of 50 ml burette into a 100 ml beaker.
- Immerse the combined glass electrode in the unknown HCl solution.
- Fill the micro burette with NaOH solution and fix it to the burette stand.
- The titration is performed by adding 1 ml of NaOH from the burette into the beaker containing 50 mL of HCl. Gently stir with a glass rod and note down the pH value when it is stable.
- Continue the addition of NaOH in 1.0 ml interval until 4 ml. pH values gradually increase with the increase in concentration of NaOH. Between 4 to 5 ml continue titration in 0.1 ml interval. Continue the titration in 1 ml upto 10 ml. At the equivalence point sudden increase in pH value will be observed.
- Plot a graph between volume of NaOH on x- axis and $\Delta\text{pH}/\Delta V$ on y-axis.

MODEL DIAGRAM · LEFT SIDE



LEFT SIDE · OBSERVATIONS & CALCULATIONS

S.No	Volume of NaOH added in ml	pH	ΔV	ΔpH	$\Delta\text{pH}/\Delta V$
1	0.0		—	—	—
2	1.0				
3	2.0				
4	2.5				
5	2.6				
6	2.7				
7	2.8				

LEFT SIDE - OBSERVATIONS & CALCULATIONS

S.No	added in ml	pH	ΔV	ΔpH	$\Delta pH/\Delta V$
8	2.9				
9	3.0				
10	3.1				
11	3.2				
12	3.3				
13	3.4				
14	3.5				
15	3.6				
16	3.7				
17	3.8				
18	3.9				
19	4.0				
20	4.1				
21	4.2				
22	4.3				
23	4.4				
24	4.5				
25	5				
26	6				
27	7				
28	8				
29	9				
30	10				

MODEL GRAPH - LEFT SIDE

$$N_1V_1 = N_2V_2$$

Where V_1 = Volume of sodium hydroxide solution = _____ ml (from graph)

N_1 = Normality of sodium hydroxide solution = _____ N (Given)

V_2 = Volume of hydrochloric acid solution = 50 ml

N_2 = Normality of hydrochloric acid solution = ?

Engineering Chemistry Lab

LEFT SIDE · CALCULATIONS

Therefore, Normality of HCl $N_2 = \frac{N_1 V_1}{V_2} =$

=

Strength of hydrochloric acid = Normality of HCl x Eq.Wt. of HCl

= $N_2 \times 36.5$

=

= g/L

Result

RESULT · RIGHT SIDE ONLY

	Reported	Given	% Error
The strength of HCl present in the given solution in g/L			

POGIL ACTIVITY

I. Title: Determination of the strength of HCl using standard NaOH by pH metry method.

II. Learning Objectives:

- i. To learn the concept of pH and the operation of pH meter.
- ii. To monitor the change in pH of a solution during the titration and to determine the equivalence point.
- iii. To learn the role of pH in determining the strength of the given solution.

III. Critical thinking Questions:

1. Glass electrode is sensitive to -----ions
a) H^+ b) OH^- c) Cl^- d) Na^+
2. What is the pH range of drinking water-----?
a) 6.5-8.5 b) 7.0 – 8.0 c) 7.5 -9.5 d) 10-12
3. Name the electrode used.
a) Glass electrode b) Platinum electrode c) Combined glass electrode.
d) silver electrode.
4. Negative logarithm of hydrogen ion concentration gives -----of the solution.
5. Electrode which can detect the concentration of a specific ion ignoring all other ions in a solution is called -----.
6. Secondary standard solution in this experiment -----.
7. Sudden increase in pH value during the addition of NaOH to HCl indicates ----

8. _____ solutions are used to calibrate pH meter.
9. What is a buffer, how many types of buffers are used?
10. pH meter is to be calibrated before starting the experiment. Why?
11. Illustrate the steps involved in the experiment?
12. What is neutralization point or equivalence point?
13. Give the significance of this experiment.

Engineering Chemistry Lab

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE · MAIN RECORD MATTER

Estimation of copper present in the given solution by colorimetric method

Aim: To estimate the amount of copper present in the given $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution by colorimetric method.

Apparatus required:

1. Cuvette
2. Beakers
3. Standard flasks
4. Colorimeter.

Reagents Required: Aqueous $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution, 1:1 ammonium hydroxide (NH_4OH) solution.

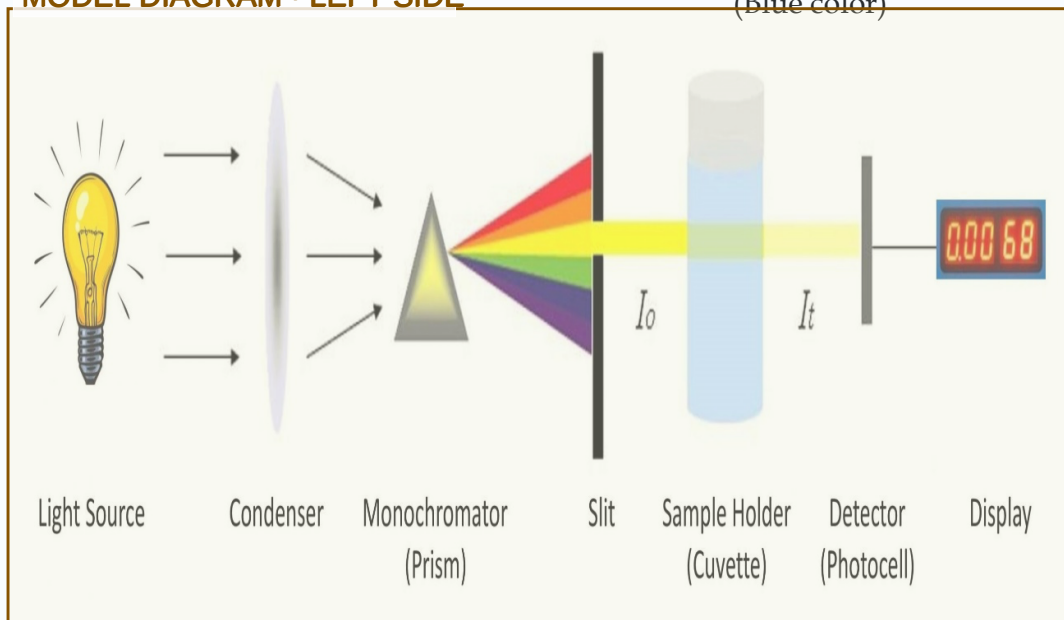
Principle:

Colorimetry is governed by Beer – Lamberts Law. According to Beer Lamberts law, absorbance (optical density) is directly proportional to the concentration of the solution and path length of absorbing medium.

Copper yields a blue color complex with aqueous ammonia. The optical density is measured in colorimeter and the amount of copper present is calculated from standard calibration graph.

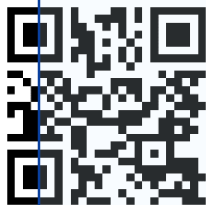


MODEL DIAGRAM · LEFT SIDE



Procedure:

RIGHT SIDE · PROCEDURE



- Take seven 100 ml standard flasks.
- Transfer $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ solution (stock solution) from the burette into the seven flasks of volumes 5ml, 10ml, 15 ml, 20 ml, 25 ml, 30 ml, 35 ml.
- Add 5 ml of 1:1 ammonium hydroxide (NH_4OH) solution to each standard flask. A blue color complex is formed.
- Make up the solution up to the mark with distilled water and shake well for uniform concentration.
- Transfer a small portion of solution from each flask into a cuvette and find the absorbance or optical density
- Plot the standard calibration graph with these values.

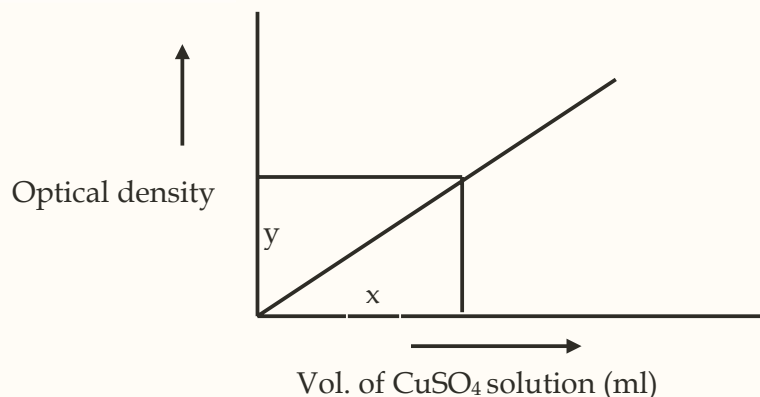
Development of color with unknown sample:

- Add 5 ml of 1:1 ammonium hydroxide (NH_4OH) solution to the given copper sulphate solution. A blue colour complex is formed.
- Make up the solution up to the mark with distilled water and shake well for uniform concentration.
- Transfer a small portion of each flask solution into a cuvette and find the absorbance or optical density.
- Plot the standard calibration graph with these values.
- Calculate the amount of copper present in the sample using the calibration graph.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

S.No	Volume of CuSO_4 solution (ml)	Absorbance or Optical Density
1	5 ml	
2	10 ml	
3	15 ml	
4	20 ml	
5	25 ml	
6	30 ml	
7	35 ml	
8	Unknown volume (x) ml	

LEFT SIDE · CALCULATIONS



Optical density of sample solution $y =$

Corresponding volume of CuSO_4 solution from graph (X ml) =

1 ml of stock solution contains 8mg/10 mg/12mg of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Amount of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ present in given solution (W_1) = $X \times 8$ (or) 10 (or) 12 mg

=

249.7 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ contains 63.5 g of copper.

Amount of Copper present in given solution (W_2) = $W_1 \times (63.5/249.7)$

=

PRECAUTIONS · NOT REQUIRED

Precautions:

1. Take only small portions of the samples into the cuvette.

2. Take the absorbance values immediately after the formation of the complex

3. The cuvette should be dried or wiped with tissue paper before inserting into the instrument.

RESULT · RIGHT SIDE ONLY

Result:

	Reported	Given	% Error
Amount of copper present in the given CuSO_4 solution			

Engineering Chemistry Lab

POGIL ACTIVITY

I. Title: Estimation of Copper by Colorimetric method

II. Learning Objectives:

- i. To determine the relation between optical density and the amount of colour developed in the solution.
- ii. To verify Beer – Lambert's law

MODEL GRAPH · LEFT SIDE

III. Critical Thinking Questions:

1. The absorbance of a solution is proportional to
a) Wavelength b) Concentration c) Path length d) b&c
2. The colour of the complex formed is
a) Blue b) Wine Red c) Reddish Brown d) Red
3. Blank solution has zero absorbance because
a) It contains some ions b) It has no ions and no colour
c) It has dissolved salts d) It has no dissolved salts
4. The filter number used in the estimation of copper is
5. is the blank solution used in colorimetry?
6. The principle of colorimetry is to verify
7. The complexing agent used in the copper estimation is
8. How colorimetry is useful in the real world?
9. When a solution of same concentration and volume are taken in a test tube and a beaker which will have more absorbance?
10. State Beer-Lambert's law?
11. What is a calibration curve? What is its significance?
12. Name the blue colour complex formed and mention the ligands present in it.
13. How is this experiment useful in the clinical laboratories?
14. Explain the use of this experiment in textile industry.

IV. Research Applications:

What is the information obtained by performing this experiment and its utility in various industries? (200 words)

Hints:

1. Colorimetry, the science of quantitatively describing color, is essential for color reproduction technology.
2. Effect of colour on absorbance in food, optical and dyeing industries.

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE - MAIN RECORD MATTER

Determination of Viscosity of Sample Oil by Redwood Viscometer No -1

Aim:

To determine the viscosity in Redwood seconds of the given sample oil at various temperatures (starting from room temperature) using Redwood Viscometer No: 1 and to check the variation of kinematic and dynamic viscosity with temperature.

Apparatus required:

1. Redwood Viscometer No: 1
2. Stopwatch
3. Thermometer
4. Kohlrausch flask
5. Weighing balance

Theory:

Viscosity is the property of a fluid by virtue of which it offers resistance to the relative motion of its own layers (i.e., resistance to flow). This method determines the viscosity of oil in terms of the time of flow (in seconds) through the Redwood Viscometer. The method is primarily intended for the determination of the viscosity of petroleum products.

Dynamic (absolute) viscosity of a fluid is defined as the tangential force per unit area on either of two parallel planes at unit distance apart, when the space between them is filled with the fluid and one plane moves relative to the other with unit velocity in its own plane. The unit of dynamic viscosity is Ns/m^2 . Kinematic viscosity of a fluid is equal to the ratio of its dynamic viscosity to its density at that temperature. The unit of kinematic viscosity is m^2/s .

Description:

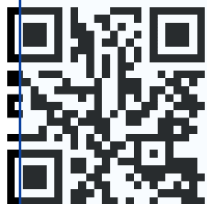
Redwood Viscometer No. 1 is suitable for measuring the viscosity of oils with flow time less than 2000 seconds (for more viscous oils, Redwood Viscometer No. 2, with a larger orifice, is used). The Redwood Viscometer works on the principle of laminar flow of liquid through a capillary (orifice) of standard dimensions under a falling head. The viscometer consists of a vertical cylinder with an orifice at the center of its base called the agate jet of diameter 1.62 mm. This cylinder is surrounded by a water bath, which maintains the test liquid at the required constant temperature; the bath is heated by an electric heater.

The cylinder is filled up to a fixed level (index mark) with the fluid whose viscosity is to be determined, and this fluid is brought to the desired temperature via the surrounding water bath. The orifice is then opened, and the time required for 50 cc (50 mL) of the oil to flow out is recorded using a stopwatch. By repeating this procedure at different bath temperatures, the variation of viscosity with temperature can be studied.

RIGHT SIDE · PROCEDURE

Procedure:

- Level the instrument with the help of circular bubble at leveling front screws.
- Fill the water bath with water.
- Close the orifice with the ball valve and fill the cylinder up to the index mark with oil through a filter or metal gauge.
- Weigh the empty 50ml Kohlrausch flask and place it below the jet.
- Before switching on the electric supply, at room temperature, note down the time taken in seconds for collecting 50cc of oil into the Kohlrausch flask with a stopwatch, by lifting the ball valve.
- Heat the viscometer bath to a few degrees above the desired temperature and as the oil attains that temperature lift the ball valve and note down the time taken for collecting 50cc of oil.
- Weigh the Kohlrausch flask with oil and calculate the mass of the oil.
- Repeat the procedure for different temperatures at regular intervals say ~~10°C say 40°C, 50°C, 60°C.~~

LEFT SIDE · CALCULATIONS (v), density (ρ) and Absolute viscosity (μ) at each temperature:

*Values of constants, $A = 26.0 \times 10^{-8}$, $B = 1.72 \times 10^{-4}$

1) At Room Temperature ____ °C

Weight of empty Kohlrausch flask (W_1) =

Weight of Kohlrausch flask with Oil (W_2) =

Weight of the Oil ($W_2 - W_1$) =

Time taken to collect 50 ml Oil (t) =

Density of Oil (ρ) = $\frac{\text{Mass of the Oil (m)}}{\text{Volume of the Oil (V)}} =$

Kinematic Viscosity (γ) = $A t - \frac{B}{t} =$
=

Absolute Viscosity (μ) = Kinematic Viscosity (γ) X Density of Oil (ρ)

LEFT SIDE · CALCULATIONS

2) At ____ °C

Weight of empty Kohlrausch flask (W₁)=

Weight of Kohlrausch flask with Oil (W₂)=

Weight of the Oil (W₂-W₁)=

Time taken to collect 50 ml Oil (t) =

Density of Oil (ρ) = $\frac{\text{Mass of the Oil (m)}}{\text{Volume of the Oil (V)}}$ =

Kinematic Viscosity (γ) = $At - \frac{B}{t}$ =
=

Absolute Viscosity (μ) = Kinematic Viscosity (γ) X Density of Oil (ρ)
=
=

3) At ____ °C

Weight of empty Kohlrausch flask (W₁)=

Weight of Kohlrausch flask with Oil (W₂)=

Weight of the Oil (W₂-W₁)=

Time taken to collect 50 ml Oil (t) =

Density of Oil (ρ) = $\frac{\text{Mass of the Oil (m)}}{\text{Volume of the Oil (V)}}$ =

Kinematic Viscosity (γ) = $At - \frac{B}{t}$ =
=

Absolute Viscosity (μ) = Kinematic Viscosity (γ) X Density of Oil (ρ)
=
=

Result:

RESULT · RIGHT SIDE ONLY

S.No ·	Oil Temperature °C	Time for collecting 50cc oil (seconds)	Kinematic Viscosity $\gamma = At - B/t$ m ² /sec.	Density ($\rho = m/v$) Kg/m ³	Absolute Viscosity N- sec/m ² ($\mu = \gamma \cdot \rho$)
1					
2					
3					

LEFT SIDE · OBSERVATIONS

Engineering Chemistry Lab

POGIL ACTIVITY

I. Title: Determination of Viscosity of oil by Redwood Viscometer No-I

II. Learning Objectives:

- i. To determine the viscosity and how it varies relative to temperature.
- ii. To understand the behavior and characteristics of engineering fluids and their applications.

III. Critical thinking Questions:

1. Viscosity index is the variation of viscosity with
 - a) Pressure b) Velocity c) Temperature d) Force
2. Units of dynamic viscosity
 - a) m^2/sec b) m^3/sec c) $\text{N-sec} / \text{m}^3$ d) $\text{N-sec}/ \text{m}^2$
3. The property of a liquid by virtue of which it resists its own flow is called
 - a) Pour point b) Viscosity c) Insulators d) Flash point
4. For a good lubricant, viscosity index should be
 - a) High b) Low c) Normal d) Unpredictable
5. oils are used for determining the Viscosity in Redwood Viscometer-1
6. Viscosity of liquids..... with increasing temperature.
7. High molecular weight oils possess viscosity.
8. A specially shaped flask for receiving the oil from jet outlet of a Redwood Viscometer is
9. The dimensions of the orifice are
10. The values of constants A & B are.....
11. Why the oil is not heated directly?
12. What are the differences between Redwood viscometer No.1 and 2?
13. How the viscosity index of lubricating oil can be improved?
14. What is the significance of viscosity determination in real world?
15. Why does viscosity change with temperature?

POGIL ACTIVITY

Engineering Chemistry Lab

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE - MAIN RECORD MATTER

Conductometric Titration of Mixture of Strong Acid and Weak Acid by Strong Base

Aim: To estimate the strength of mixture of strong acid and weak acid conductometrically using standard NaOH solution.

Apparatus required:

1. Conductivity meter
2. Conductivity cell
3. 10 ml micro burette
4. 250 ml conical flask
5. 100 ml beaker

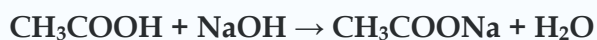
Reagents required:

1. Oxalic acid
2. NaOH solution
3. HCl solution, Acetic acid solution
4. Phenolphthalein Indicator

Principle:

The determination of equivalence point of a titration by this method is based upon the measurement of the conductance during titration. The conductance varies before and after the equivalence point. The reason for this is electrical conductance of solution depends upon the number of ions and their mobilities.

The titration of mixture of two acids in which one is weak (acetic acid) and other one strong (HCl), by a strong base (NaOH) can be carried out conductometrically. The reactions take place are as follows



In this conductometric titration there are two break points. The first break point corresponds to the neutralization of strong acid. Since HCl is strong acid, it will dissociate completely and hence due to common ion effect it will also suppress the dissociation of acetic acid. Therefore, HCl will get titrated first followed by acetic acid. When the strong acid has been completely neutralized only then the weak acid starts neutralizing. The second break point corresponds to the neutralization of weak acid and after that the conductance increases due to the excess of OH^- ions.

Step-1: Preparation of standard Oxalic acid solution



- Weigh accurately about 3.2 g of the given oxalic acid salt using the weighing balance and transfer into a clean 100 ml volumetric flask.
- Dissolve the transferred salt in distilled water and make up the solution up to the mark.
- Shake the volumetric flask for a while to get uniform concentration.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Weight of Oxalic acid salt + weighing bottle (W_1) =

Weight of empty weighing bottle (W_2) =

Weight of Oxalic acid salt transferred ($W_1 - W_2$) =

$$\begin{aligned} \text{Normality of Oxalic acid salt solution} &= \frac{\text{Weight of salt}}{\text{Equivalent weight of salt}} \times \frac{1000}{100} \\ &= \\ &= \end{aligned}$$

(Equivalent Weight of Oxalic acid is 63)

Step-2: Standardization of NaOH solution



- Pipette out 20 ml of oxalic acid solution into a 250 ml conical flask.
- Add one or two drops of phenolphthalein indicator.
- Fill the burette with sodium hydroxide adjust to zero, then fix it to the burette stand.
- Now titrate the above oxalic acid solution against NaOH solution till the colourless solution turns to pale pink which is the end point of the titration.
- Note down the burette reading repeat the titration until concurrent readings are obtained.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Observations and Calculations:

S. No.	Volume of oxalic acid (V ₁ ml)	Burette reading(ml)		Volume of NaOH solution run down (v ₂ ml)
		Initial	Final	
1.	20			
2.	20			
3.	20			
4.	20			

$$V_1N_1 = V_2N_2$$

Where V₁ -Volume of oxalic acid solution =

N₁ - Normality of oxalic acid solution =

V₂ - Volume of sodium hydroxide solution =

N₂ - Normality of sodium hydroxide solution =?

$$N_2 = \frac{N_1V_1}{V_2}$$

=

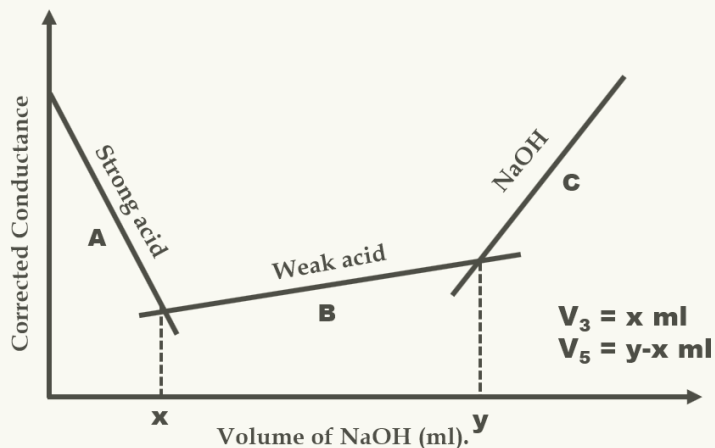
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RIGHT SIDE · PROCEDURE

Step-3: Estimation of Mixture of acids (HCl + CH₃COOH)

- Make up the given unknown (HCl+CH₃COOH) mixture solution up to the mark with distilled water.
- Transfer 50 ml of above solution with the help of 50 ml burette into a 100 ml beaker.
- Immerse the conductivity cell in the mixture solution.
- Fill the micro burette with standardized NaOH solution and fix it to the stand.
- Add 0.5 ml of NaOH from the micro burette into the beaker, stir well and note the conductance.
- Continue the addition of NaOH up to 12ml. Conductance first decreases by the addition of NaOH and increases slowly after complete neutralization of HCl, a sharp steep in increase of conductance is observed after neutralization of CH₃COOH.
- Draw a graph plotting the volume of NaOH on X-axis and conductance on Y-axis. The point of intersection between lines **A** and **B** is the end point that corresponds

to the total volume of NaOH (**x**) required for the neutralization of HCl. The point of intersection between lines **B** and **C** is the end point that corresponds to the total volume of NaOH (**y**) required for the complete neutralization of mixture of HCl and CH₃COOH.

MODEL DIAGRAM · LEFT SIDE

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Observations and Calculations:

	Volume of NaOH (V)	Conductance in mS At 25°C (k)	Corrected Conductance (k ¹) $\left(\frac{50+V}{50} \right) * k$
MODEL GRAPH · LEFT SIDE			
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			

LEFT SIDE - CALCULATIONS**Determination of strength of strong acid (HCl):**

$$V_3 N_3 = V_4 N_4$$

Where

 V_3 - Volume of sodium hydroxide solution = _____ ml ("x" from graph)

 N_3 - Normality of sodium hydroxide solution = _____ N (Given/step-2)

 V_4 - Volume of mixture of acids = 50 ml

 N_4 - Normality of hydrochloric acid = ?

$$\text{Therefore, } N_4 = \frac{V_3 N_3}{V_4}$$

Strength of hydrochloric acid (d_1)

= Normality of HCl solution x Eq.Wt. of HCl

= $N_4 \times 36.5$

=

Determination of strength of weak acid (CH_3COOH):

$$V_5 N_5 = V_6 N_6$$

Where

 V_5 - Volume of sodium hydroxide solution = _____ ml ("y-x" from graph)

 N_5 - Normality of sodium hydroxide solution = _____ N (Given/step-2)

 V_6 - Volume of mixture of acids = 50 ml

 N_6 - Normality of CH_3COOH = ?

$$\text{Therefore, } N_6 = \frac{V_5 N_5}{V_6}$$

Strength of acetic acid (d_2)= Normality of CH_3COOH solution x Eq.Wt. of CH_3COOH = $N_6 \times 60$

=

Precautions:**PRECAUTIONS · NOT REQUIRED**

1. The electrodes must be immersed properly.
2. The temperature of the solution should not change during the measurement.

Result:**RESULT · RIGHT SIDE ONLY**

	Reported	Given	% Error
The strength of HCl present in the solution (d_1)			
The strength of CH_3COOH present in the solution (d_2)			

Engineering Chemistry Lab**POGIL ACTIVITY**

I. Title: Estimate of strength of mixture of Strong Acid and Weak Acid conductometrically using standard NaOH solution.

II. Learning Objectives:

- i. To learn the concept of conductance and the operation of Conductivity meter.
- ii. To monitor the total conductance of a solution during the titration and to determine the equivalence point.
- iii. To learn the role of conductance in determining the strength of the given solution.

III. Critical thinking Questions:

1. Units of conductance are
a) Ohms b) Moles / L c) Siemens d) cm^{-1}
2. Conductance depends upon
a) Temperature b) Mobility c) Concentration d) All
3. The shape of the graph obtained
a) X-shape b) V-shape c) Y-shape d) boat shape
4. The indicator used in the standardization of sodium hydroxide is
a) Methyl orange b) EBT c) Phenolphthalein d) Methyl red
5. The titrations in which endpoint is determined by conductance measurements are called _____
6. The platinum strips in a conductivity cell are coated with platinum black for _____
7. Calibration of an instrument is done to get _____
8. The conductance of the acid volume before endpoint. _____
9. Solutions used for calibration of conductivity meter is _____ .
10. Why is AC current used in conductometric titration?
11. How temperature effects this titration?
12. Why the conductance range should be changed during the calibration?

IV. Research Applications:

What is the information obtained by performing this experiment and its utility in various fields? (200 words)

Hints:

1. How conductance measurements are useful in chemical and biological fields?
2. Use of conductance measurements in various industries.

POGIL ACTIVITY

Engineering Chemistry Lab

POGIL ACTIVITY

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE · MAIN RECORD MATTER

Determination of surface tension of a liquid by drop method using Stalagmometer

Aim: To determine the surface tension of a given liquid at room temp using stalagmometer by drop number method.

Apparatus required:

1. Stalagmometer
2. Beaker
3. Rubber tubes

Reagents required:

1. Acetone
2. Propanol
3. Distilled water

Theory:

Surface tension is measured as the force acting normally per unit length on an imaginary line drawn on the free liquid surface at rest. It is represented by the symbol γ (or S) and units are Nm^{-1} or dyne-cm^{-1} . Surface tension is based on the molecular theory of matter. According to this theory, cohesive forces among liquid molecules are responsible for the phenomenon of surface tension. The molecules well inside the liquid are attracted equally in all directions by the other molecules. The molecules on the surface experience an inward pull. So, a network is formed against the inward pull, to move a molecule to the liquid surface. It results in a greater potential energy on surface molecules. To attain minimum potential energy and hence stable equilibrium, the free surface of the liquid tends to have the minimum surface area and thereby it behaves like a stretched membrane.

Example of the property of surface tension,

- (a) The ability of some objects to float on the surface of water, even though they are denser than water.
- (b) Surface tension is also seen in the ability of some insects, such as water striders, and even reptiles like basilisk, to run on the water's surface.

Principle:

Surface tension is the result of force of attraction between the molecules present on the surface of a liquid. It is the property along the unit length on the surface of a liquid. The formation of a drop depends upon the surface tension of a liquid and all the liquids will not give same number of drops per same volume of liquid.

The surface tension of a given liquid can be calculated by the formula

$$\frac{\gamma_1}{\gamma_2} = \frac{n_2 d_1}{n_1 d_2}$$

γ_1 =surface tension of water (72.8 dyne/cm)

γ_2 =surface tension of the given liquid

n_1 =no. of drops of water

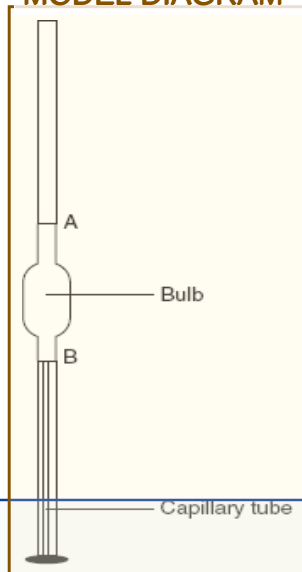
n_2 = no. of drops of given liquid

d_1 = density of water (1 gr/cc)

d_2 = density of given liquid

Stalagmometer description:

Stalagmometer is a device for determining the number of drops in each volume of liquid and in turn used in calculating the surface tension. It is also called a stactometer or stalogmometer. The device is a capillary glass tube whose middle section is widened. The middle section which is blubbed is marked with an upper mark(A) and lower mark(B). The lower end of the tube is narrowed to force the fluid to fall out of the tube as a drop.



RIGHT SIDE - PROCEDURE

Procedure:



- i. Take a clean and dry weighing bottle and weigh it (w_1).
- ii. Into a clean Stalagmometer suck distilled water till the water level rises above the upper mark A.
- iii. Clamp the Stalagmometer and allow the water drops to fall into the weighing bottle.
- iv. Start counting the number of drops (n_1) when the meniscus crosses the mark A and stop counting when it passes the mark B.
- v. Take the weight of the weighing bottle with the collected water (w_2).
- vi. Repeat the procedure for the second time and take the average values of n_1 and w_2 .
- vii. The experiment is repeated with the given 5% acetone and 10% acetone solutions two times each and take the average values of n_2 and w_3 .
- viii. The density of the liquid sample is calculated (d_2), from which the surface tension of the liquid sample can be calculated.

Observations

LEFT SIDE - OBSERVATIONS		No. of drops		Average	Surface tension (dynes/cm)
		Trial-1	Trial-2		
1	Water			$n_1 =$	
2	5% Acetone solution			$n_2 =$	
3	10% Acetone solution			$n_3 =$	

LEFT SIDE · OBSERVATIONS & CALCULATIONS

Room temperature = t °C

- Weight of empty bottle (w_1) =
- Weight of bottle + water (w_2) =
- Weight of bottle + 5% Acetone (w_3) =
- Weight of bottle + 10% Acetone (w_4) =
- Weight of water ($w_2 - w_1$) =
- Weight of 5% Acetone ($w_3 - w_1$) =
- Weight of 10% Acetone ($w_4 - w_1$) =

Calculations:**1. Calculation of Density**Density of water = d_w or $d_1 = 1 \text{ gr/cc}$

$$\text{Density of the 5\% Acetone } (d_2) = \frac{(w_3 - w_1)}{(w_2 - w_1)} \times d_w$$

=

=

$$\text{Density of the 10\% Acetone } (d_3) = \frac{(w_4 - w_1)}{(w_2 - w_1)} \times d_w$$

=

=

LEFT SIDE · CALCULATIONS

2. Calculation of Surface Tension

Surface Tension of 5% Acetone

$$\frac{\gamma_1}{\gamma_2} = \frac{n_2 d_1}{n_1 d_2}$$

γ_1 = Surface Tension of water
= 72.8 dyne/cm

γ_2 = Surface Tension of 5% Acetone = ?

n_1 = Number of water drops =

n_2 = Number of 5% Acetone =

d_1 = density of water (1 gr/cc)

d_2 = density of given liquid =

Surface Tension of 10% Acetone

$$\frac{\gamma_1}{\gamma_3} = \frac{n_3 d_1}{n_1 d_3}$$

γ_1 = Surface Tension of water
= 72.8 dyne/cm

γ_2 = Surface Tension of 5% Acetone = ?

n_1 = Number of water drops =

n_3 = Number of 10% Acetone =

d_1 = density of water (1 gr/cc)

d_3 = density of given liquid =

RESULT · RIGHT SIDE ONLY

Result: -

Surface tension of given liquids	Reported	Given	% Error
5% Acetone solution			
10% Acetone solution			

Engineering Chemistry Lab**POGIL ACTIVITY**

- I. Title :** To determine the surface tension of a given liquid at room temp using stalgmometer by drop number method.

II. Learning Objectives

- The combination of adhesive and cohesive forces causes water to rise in a thin tube (capillary action).
- Correlate the property of surface tension with different natural phenomena.
- Practice using scientific notation and calculating experimental error.
- The role of capillary action in measuring surface tension.

III. Critical thinking Questions

- The unit of surface tension in CGS system is
a) N/m b) Kg/cm c) Dynes/cm d) None
- The apparatus used for the determination of surface tension of a liquid is
a) Odometer b) Stalagmometer c) Consolidometer d) Viscometer
- Real life examples of surface tension
a) Water leaking from tap b) Insects walking on water
c) Floating needle d) All
- The most common method for measuring surface tension is
a) Drop count method b) Drop weight method
c) Both d) Only a
- Drop of water form a spherical shape on a flat surface due to
- On increasing the temperature, the surface tension of the liquid _____.
a) Viscosity b) Conductivity c) Absorption d) Surface
- The rise of liquid in a capillary tube does not depend on
a) Angle of contact b) Density of the liquid
c) Radius of the capillary tube d) Atmospheric pressure
- The work done per unit area in increasing the surface area of a liquid under isothermal condition is _____.
- Plants get water through roots because of _____.
- At the boiling point, of water its surface tension.
- What are the precautions to be taken during the operation of stalagmometer?
- What is the principle involved in the usage of stalagmometer?
- Why the floating of wood on the surface of water is not an example of capillary action?
- Suggest three more experimental methods for determination of surface tension.

IV. Research Applications

What is the information obtained by performing this experiment and its applications in various fields.

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE - MAIN RECORD MATTER

Estimation of acid value of given lubricant oil.

Aim: To estimate the acid value of the given lubricant oils

Apparatus required:

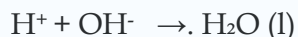
1. 50 ml burette
2. 250 ml conical flask
3. 100 ml beaker

Reagents required:

1. Oxalic acid
2. KOH solution
3. Sample oils
4. Isopropyl alcohol,
5. Phenolphthalein indicator.

Principle:

The acid value of lubricating oil is defined as the number of milligrams of potassium hydroxide required to neutralize the free acid present in 1g of the sample oil. In good lubricating oils, the acid value should be minimum. Increase in acid value should be taken as an indicator of oxidation of the oil which may lead to gum and sludge formation besides corrosion. A known weight of the sample oil is dissolved in a suitable solvent and titrated with a standard alcoholic solution of KOH to a definite end point.



Procedure:

Step -1: Preparation of 0.005N oxalic acid solution:

- Weigh accurately the given oxalic acid salt and transfer it into a clean 100 ml volumetric flask.
- Dissolve it in distilled water and make up the solution up to the mark.

LEFT SIDE - OBSERVATIONS & CALCULATIONS

Weight of the bottle + oxalic acid (W_1) = _____

Weight of empty weighing bottle (W_2) = _____

Weight of oxalic acid transferred into 100 ml volumetric flask ($W_1 - W_2$) = _

LEFT SIDE · CALCULATIONS

$$N_1 - \text{Normality of oxalic acid solution} = \frac{\text{Weight of oxalic acid}}{\text{Equivalent weight}} \times \frac{1000}{100}$$

$$=$$

$$= \underline{\hspace{2cm}}$$

* Equivalent weight of Oxalic acid is 63

RIGHT SIDE · PROCEDURE

Step-2: Standardization of potassium hydroxide solution:

- Pipette out 20 ml of oxalic acid solution into a conical flask.
- Add one or two drops of phenolphthalein indicator and then titrate against potassium hydroxide taken in a burette.
- The end point is colorless to pale pink.
- Repeat the titration until concurrent readings are obtained.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

S. No.	Volume of oxalic acid (V ₁ ml)	Burette reading(ml)		Volume of KOH solution run down (v ₂ ml)
		Initial	Final	
1.	20			
2.	20			
3.	20			

$$V_1 N_1 = V_2 N_2$$

Where V₁ - Volume of oxalic acid solution =

N₁ - Normality of oxalic acid solution =

V₂ - Volume of potassium hydroxide solution =

N₂ - Normality of potassium hydroxide solution = ?

$$N_2 = \frac{N_1 V_1}{V_2}$$

=

= _____ N

Step-3: Determination of acid value of the given sample oils

RIGHT SIDE · PROCEDURE

- Weigh out accurately about 1 g of the oil under test into a 250 ml conical flask.
- Add 10 ml of neutral alcohol and heat the flask over hot water bath for 30 minutes.
- Cool the flask and the contents to room temperature and add a few drops of phenolphthalein indicator.
- Titrate with the standard 0.01N KOH solution until a faint permanent pink colour appears at the end point.
- Repeat the same process for the given sample oils.

LEFT SIDE · OBSERVATIONS & CALCULATIONS

S.No	Weight of Oil sample taken	Burette reading(ml)		Volume of KOH solution run down (ml)
		Initial	Final	
Sample oil -1				
Sample oil-2				

a) Sample oil -1

$$\text{Acid Value of sample oil -1} = \frac{\text{Normality of KOH} \times \text{Volume of KOH} \times 56}{\text{Weight of the oil taken (gm)}}$$

Where 56 represents the amount of KOH in mg present per each ml of 1N KOH solution (1000 ml of 1 N KOH=56 g of KOH)

=

b) Sample oil -2

$$\text{Acid Value of sample oil -2} = \frac{\text{Normality of KOH} \times \text{Volume of KOH} \times 56}{\text{Weight of the oil taken (gm)}}$$

=

Result:

1. Acid value of sample oil -1 =
2. Acid value of sample oil -2 =

RESULT · RIGHT SIDE ONLY

Precautions

PRECAUTIONS · NOT REQUIRED

1. The alcoholic solution of KOH should be freshly prepared and should be standardized just before the test.

POGIL ACTIVITY

I. Title: Estimation of acid value of the given lubricant oils.

II. Learning Objectives:

- i. To estimate the acid value of the given lubricant oils.
- ii. To understand the effect of acid value on quality of lubricants.

III. Critical Thinking Questions:

1. Which indicator is used in the experiment?
a) phenolphthalein b) methyl orange c) litmus indicator d) bromothymol blue
2. What is the acid value of the groundnut oil?
a) 0.2-8.0 b) 0.4-0.8 c) 10.0-35.2 d) 0.4-2.2
3. What is the saponification oil of coconut oil?
a) 194-196 b) 201-203 c) 253-260 d) 194-197
4. Which of the following has no acid value?
a) Cotton seed oil b) Rape seed oil c) Whale oil d) Hard oil
5. What is the unit of acid value?
6. Define acid value of lubricating oil?
7. Give the principle involved in acid value?
8. What is the significance of lubricant oil?
9. Which oil has high acid value?
10. What is the role of isopropyl alcohol?
11. When will the acid value be high?
12. How do you determine the acid value of given lubricant oil?
13. Why acid value determination is important for lubricating oil?

POGIL ACTIVITY

Engineering Chemistry Lab

POGIL ACTIVITY

Preparation of Phenol-Formaldehyde Resin (Bakelite) RIGHT SIDE · MAIN RECORD MATTER

Aim: To prepare Phenol formaldehyde (Bakelite) resin.

Apparatus Required:

1. 250 ml beaker
2. 100 ml beaker
3. 5 ml measuring jar
4. Glass rod
5. Filter paper

Reagents required:

1. Phenol (2g)
2. 40% aqueous formaldehyde solution or formalin (2.5 mL),
3. Glacial acetic acid (5 mL)
4. Con HCl (8mL).

Principle:

Phenolic resins or phenoplast are condensation polymerization products of phenolic derivatives like phenol with aldehydes like formaldehyde. The polymers which on heating change irreversibly into hard rigid and infusible materials are called thermosetting polymers. These polymers are usually prepared by heating relatively low molecular mass, semi fluid polymers, which becomes infusible and form an insoluble hard mass on heating. The hardening on heating is due to the formation of extensive cross-linking between different polymeric chains. This led to the formation of a 3-Dimensional network of bonds connecting the polymer chains.

Following steps are involved in the preparation of Bakelite:

Step 1: Formation of Methylol Phenol Derivative

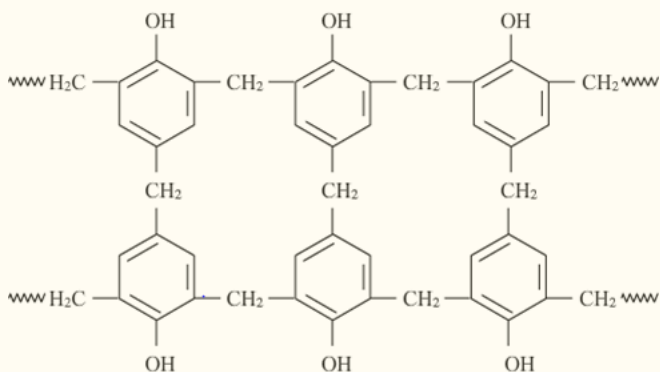
Initially the monomers combine to form methylol phenol derivative depending upon phenol to formaldehyde ratio.

Step 2: Formation of Novolac

The phenol formaldehyde derivatives react among themselves or with phenol to give a linear polymer or a higher cross-linked polymer.

Step 3: Formation of Bakelite by condensation

A highly cross-linked thermosetting polymer called bakelite is formed by further condensation of novolac by an addition of Hexamethylene tetra amine which develops - CH₂ - cross links.



Bakelite

Procedure:**RIGHT SIDE · PROCEDURE**

- Place 5 mL of glacial acetic acid and 2.5 mL of 40 % aqueous formaldehyde solution in a 100 ml beaker. Add 2 g phenol safely.
- Wrap the beaker with a wet cloth or place it in a 250 mL beaker having small amount of water in it.
- Add 1ml of Con HCl drop wise with vigorous stirring by a glass rod till the polymerization point is reached, a white precipitate will form and dissolve.
- At the point where polymerization begins, the white precipitate will not dissolve. Continue to stir as the plastic forms and becomes pink color.
- Wash the pink residue several times with distilled water to make it free from acid.
- Filter the product and weigh it after drying in folds of a filter paper or in an oven. Report the yield of polymer formed.

**LEFT SIDE · OBSERVATION / WEIGHING**

Weight of empty watch glass (W_1)	=	g
Weight of watch glass + polymer formed (W_2)	=	g
Weight of polymer formed ($W_2 - W_1$)	=	g

Result**RESULT · RIGHT SIDE ONLY**

Yield of phenol-formaldehyde resin prepared is-----gm.

Engineering Chemistry Lab**POGIL ACTIVITY**

I. Title: Preparation of Phenol formaldehyde resin.

II. Learning Objectives:

- i. To understand the mechanism of condensation reaction.
- ii. To gain knowledge about steps involved in the preparation of Bakelite.

III. Critical thinking Questions:

1. Bakelite is a
 - a) Thermoplastic
 - b) Thermoelastic polymer
 - c) Thermosetting
 - d) None of these
2. The condensation product of phenol and formaldehyde is known as
 - a) Phenolic resins
 - b) Polyvinyl chloride
 - c) Urea formaldehyde resin
 - d) Polystyrene
3. Phenol formaldehyde is also called as_____.
4. _____colour indicates the formation of Bakelite.
5. Number of washings is given to the polymer to remove_____.
6. _____is added in the preparation of Bakelite to develop cross links.
7. _____ is the linear polymer formed as Stage. A resin in the preparation of Bakelite.
8. _____and_____are the monomers of Bakelite.
9. What is condensation polymerization?
10. Why Bakelite is a thermosetting resin?
11. What is the functionality of the monomers in Bakelite?
12. Give the applications of Bakelite in electrical field?
13. Bakelite is used in the purification of drinking water? Give reason.

POGIL ACTIVITY

Experiment no:

Date:

RIGHT SIDE · MAIN RECORD MATTER

Preparation of Nylon-6,6

Aim: To synthesize Nylon-6,6 polymer by condensation polymerization of adipoyl chloride and hexamethylenediamine.

Apparatus required:

1. 50 ml beaker
2. 150 ml beaker
3. 10 ml measuring jar
4. Glass rod or forceps
5. Filter paper

Reagents required:

1. Adipoyl chloride
2. 5% hexane -1,6- diamine
3. Cyclohexane
4. 20% sodium hydroxide

Principle:

Nylon-6,6 is a type of synthetic polymer that belongs to the family of polyamides. It is prepared by the condensation polymerization (also called step-growth polymerization) of two different monomers such as hexamethylenediamine (a diamine with 6 carbon atoms) and adipic acid (a dicarboxylic acid with 6 carbon atoms).

The polymerization occurs by a condensation reaction between the amine (-NH_2) groups of hexamethylenediamine and the carboxyl (-COOH) groups of adipic acid.

Each amine group reacts with a carboxyl group to form an amide bond (-CONH-), releasing a molecule of water (H_2O) as a byproduct.

This reaction repeats, linking many monomers to form a long polymer chain called Nylon-6,6.

The polymer is prepared from the diamine and the dicarboxylic acid.

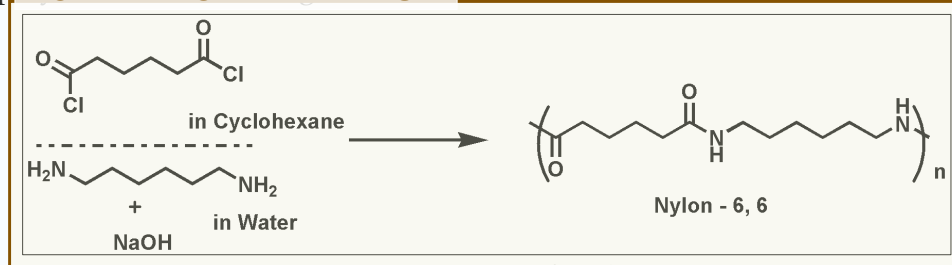


Figure: Formation of Nylon-6,6

Procedure:**RIGHT SIDE · PROCEDURE****1. Preparation of Solutions**

- Prepare 10 mL of 5% adipoyl chloride in cyclohexane in a 50-mL beaker.
- Prepare 10 mL of 5% aqueous hexamethylenediamine in a 150-mL beaker.
- Add 7 drops of 20% sodium hydroxide solution to the diamine solution.

2. Layer Formation

Tilt the diamine solution and slowly pour the adipoyl chloride solution down the wall of the beaker to form two immiscible layers.

3. Polymer Formation

- A nylon film forms at the interface of the two layers.
- Using forceps or a copper wire hook, grasp the film and slowly pull to form a continuous nylon fiber (rope).

4. Bulk Polymerization

- Stir the remaining solution to promote bulk polymer formation that settles at the bottom.

5. Purification

- Rinse both the formed rope and the bulk polymer in water three times.
- Remove excess water by sandwiching the fiber between filter paper.

6. Yield Calculation

- Weigh the dried fiber using an analytical balance and note the result.

Result:**RESULT · RIGHT SIDE ONLY**

Yield of Nylon 6,6 prepared is----- gm.

POGIL ACTIVITY

1. What are the two monomers used in the synthesis of Nylon 6:6?
 - a) Caprolactam and Terephthalic acid
 - b) Adipic acid and Hexamethylenediamine
 - c) Sebacic acid and Ethylene glycol
 - d) Acetone and Ethylene oxide
2. The numbers "6,6" in Nylon 6:6 refer to:
 - a) Number of polymer chains
 - b) Number of amide units per chain
 - c) Number of carbon atoms in each monomer
 - d) Polymer crystallinity index
3. Which of the following is a key property of Nylon 6:6?
 - a) High melting point
 - b) Poor tensile strength
 - c) High electrical conductivity
 - d) Solubility in water
4. Nylon 6:6 is used in which of the following applications?
 - a) Food additives
 - b) Textile fibers and automotive components
 - c) Paint thinners
 - d) Perfumes
5. How does Nylon 6,6 differ from Nylon 6 in thermal stability
6. Why does Nylon 6,6 show better wear resistance than many other polymers
7. How will Nylon 6,6 performance change in humid environments

POGIL ACTIVITY